

EP03 — How does the internet reach your phone?

HINGLISH (Roman)

01 Title

Internet actually aapke phone tak pahunchta kaise hai? Ek spoiler jo zyada logon ko surprise karta hai: ye almost **kabhi** wireless nahi hota. Chalo aapke data ka asli safar dikhata hoon.

[CUE] Title pe ~2s ruko. Curious, "ek secret reveal karta hoon" tone.

02 The Hook

Aap yahaan India mein ek video tap karte ho, aur wo America mein baithe server se load hota hai — kuch **milliseconds** mein. Koi wire aapke haath ko chhuti tak nahi. Toh ye data poori duniya itni tezi se cross kaise karta hai? Chalo follow karte hain.

[CUE] Energy up. "across the world" haathon se gesture karo.

03 Aaj ka sawaal

Toh aaj ka sawaal: jab aap ek website kholte ho, **physically** travel kya karta hai — aur kitni **door** tak jaata hai? Chalo pata karte hain.

[CUE] Slow down. "kitni door" ko ek second hang hone do.

04 Idea 1 — WiFi is the last hop

Idea ek. "WiFi" word sabko dhoka deta hai. WiFi sirf **last few metres** hai. Aapka phone router se radio waves se baat karta hai — par wo wireless hop bahut chhota hai, bas deewar pe lage box tak. Router ke baad? **Koi air nahi.**

[CUE] Point: phone → router. "bas deewar pe lage box tak" pe zor.

05 Idea 2 — It's cables

Idea do. Router ke aage, sab **wires** hain — bahut saari. Fibre-optic cables aapke sheher mein, deshon ke beech mein chalti hain, aur phir giant cables **samundar ke farsh** pe padi hoti hain, continents ko jodti hui. Lagbhag ninety-nine percent internet physical cable hai. Air nahi. Satellites nahi.

[CUE] Chain trace karo: router → city → undersea → data centre.

06 Idea 3 — Finding the server

Idea teen. Aapka data randomly nahi ghoom raha — wo ek **exact computer** tak jaa raha hai. Aap ek naam type karte ho jaise youtube-dot-com, aur wo naam ek **address** pe map hota hai — ek IP, computers ke liye phone number jaisa. Aapki request us address tak jaati hai, sahi server dhoondhti hai, aur reply **wapas** aapke phone tak aata hai.

[CUE] Light rakho — packets agle episode mein detail mein.

07 Idea 4 — Speed of light

Idea chaar — ye **instant** kyun lagta hai? Kyunki fibre aapka data **light** ke pulses ki tarah le jaati hai. Mumbai se London lagbhag saat hazaar kilometre hai, aur ek round trip mein takreeban ek sau bees milliseconds lagte hain. Light universe ki sabse tez cheez hai — toh poori duniya ke paar bhi, ye turant lagta hai.

[CUE] Mumbai → London stat card reveal karo jab numbers bolo.

08 Why it matters

Aur jab ye samajh aata hai, toh bahut cheezein click hoti hain. Door wala server matlab zyada distance, matlab zyada delay — yahi **ping**, ya lag, hai aapke games mein. "Cloud" bas bade buildings mein servers hain, in cables se jude. Aur agar ek undersea cable kat jaaye, toh poore region ka internet slow ho sakta hai.

[CUE] Gaming lag se relate karo — audience ise feel karti hai.

09 Common myth

Toh myth: log sochte hain internet wireless hai — air mein float karta, satellites se beam hota. Reality: lagbhag ninety-nine percent **physical cable** hai, including undersea fibre continents ke beech. Wireless part — WiFi, mobile data — sirf aap tak ka **last hop** hai.

[CUE] Myth pe ek beat ke baad green "Reality" card reveal karo.

10 Recap

Quick recap. Ek: internet ek giant web of **physical cables** hai — wireless bas aap tak ke last few metres hai. Do: aapka data **light** ki tarah server ke address tak aur wapas race karta hai — yahi millisecond magic hai.

[CUE] Crisp. Do saaf lines.

11 CTA

Toh internet asal mein cables plus light hai — poori duniya ko lapeitta hua. Agar isne dimaag thoda hila diya, toh Katixo KhojLab ko subscribe karo. Agla sawaal: jis pal aap "Send" click karte ho, **actually** hota kya hai? Milte hain agle video mein.

[CUE] Warm, friendly. Subscribe line pe awaaz mein smile.
